REAL ANALYSIS II · SEMESTER 5 · MODULE 4

# Prepare: Sequences of Functions

Everything you need before you open the question paper, from Real Analysis I up to Bartle & Sherbert §8.1 (8.1.1–8.1.10) and §8.2 (8.2.1–8.2.2).

| | |
|---|---|
| **File 1 of 3** | **Prepare** (this file): learn the ideas |
| **File 2** | Model paper: past questions + new ones |
| **File 3** | Answer key: full solutions |

Each answer in File 3 starts with topic codes such as **T6** **P3** **M4** . If a code is new to you, read that topic here first.

## How to read this file

One topic at a time, in order. Every topic uses the same boxes.

**REMIND ME**

Facts from earlier (Plus Two, Real Analysis I) that you may have forgotten.

**MATHS TOOL**

A small piece of maths, taught from zero the first time it is needed.

**THE IDEA**

The meaning in plain words and pictures, before any symbols.

**DEFINITION · SAY IT WORD FOR WORD**

Bartle's exact wording. Examiners give marks for the exact words.

Underneath: what it says, in plain words.

**SKELETON**

The lines you can write **before** you have any idea. They earn most of the marks. ? marks the bit you must work out.

**SCRATCH (ROUGH WORK, NOT FOR THE EXAM)**

How we **find** the K, the point $x_n$ or the bound. Like rough work in the margin.

**PROOF (WHAT YOU WRITE)**

The clean answer, in the right order. ∎

**HYPOTHESES CHECKLIST**

☐ Each condition of a theorem
what goes wrong without it, with a counterexample
code C12

**RESULT**

What you must remember and write in the exam.

**EXAM TRAP**

A common mistake, and how to avoid it.

**CHECK YOURSELF**

**A question.**

The answer, just below. Cover it with your thumb first.

## How to study with it

> From research on how strong students learn analysis.

1. **Do, don't re-read.** Re-reading and highlighting barely help. Closing the file and writing a definition from memory helps a lot.
2. **Explain every line.** For each proof step, say *why* it is true (a definition, an earlier line, or a named theorem). The grey "why" lines show you how.
3. **Skeleton first.** In the exam, write the skeleton, leave a gap, and fill the gap last.
4. **Know the counterexamples.** Half of analysis is knowing which picture breaks a wrong idea.
5. **Little and often.** Use the recall plan on the last pages: today, in 1, 3, 7 and 14 days.

# One idea for the whole module

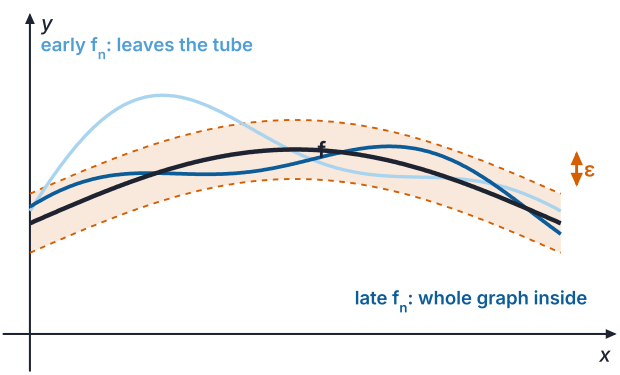
**THE E-TUBE**

A sequence of functions is a **film**: $f_1$, $f_2$, $f_3$, … are frames, each a whole graph.

**Pointwise**: stand at one x and watch the height $f_n(x)$ settle down. Every x settles, but some may take much longer than others.

**Uniform**: draw a thin tube round the final graph f. From some frame on, the **whole** graph of $f_n$ lies inside the tube, however thin the tube.

Every question in this module asks: which of these two happens, and what does it buy us?




Uniform convergence = the ε-tube picture. Draw a tube of half-width ε round the graph of f. $f_n \to f$ uniformly when, for every ε, from some n = K on **the whole graph** of $f_n$ lies inside the tube.

## Codes and map

P1–P6 Real Analysis I and Plus Two bridge

M1–M5 Maths tools

T1–T15 Module 4 topics

C12 a counterexample from the library

Q12 question 12 of the model paper

| SYLLABUS UNIT | TOPICS |
|---|---|
| 19. §8.1.1–8.1.3 Pointwise convergence | T1, T2 |
| 20. §8.1.4–8.1.10 Uniform convergence, uniform norm, Cauchy criterion | T3–T10 |
| 21. §8.2.1 Interchange of limits | T11, T12 |
| 22. §8.2.2 Interchange of limit and continuity | T13, T14 (+ T15) |

At least 10 of the 70 external marks. In past papers: the Cauchy criterion was a 10–15 mark question twice; "uniform limit of continuous functions is continuous" and the norm test were 5-mark questions; pointwise limits like $\lim e^{-nx}$ appear almost every year.

## The whole module in one story

**THE IDEA**

**1.** Fix x and let n run: that gives the **pointwise** limit f(x). Work it out case by case (T1, T2).

**2.** Ask for one K that works for **all** x at once: that is **uniform** convergence. To show it fails, find points $x_n$ where $f_n$ stays far from f (T3, T4).

**3.** Measure the worst gap with the uniform norm $\|f_n - f\|$. Uniform ⇔ worst gap → 0. Find the worst gap with calculus or a bound (T5–T9).

**4.** The Cauchy criterion tests uniform convergence without knowing f (T10).

**5.** Pointwise limits can destroy continuity and spoil integrals; uniform limits of continuous functions stay continuous (T11–T15).

### The four sequences you will meet again and again

| $F_N(X)$ | POINTWISE LIMIT | UNIFORM? |
|---|---|---|
| $x/n$ | 0 on $\mathbb{R}$ | on [0, 1] yes; on $\mathbb{R}$ no |
| $x^n$ | 0 on [0, 1), 1 at 1 | on [0, b], b < 1, yes; on [0, 1] no |
| $(x^2+nx)/n$ | x on $\mathbb{R}$ | on bounded sets yes; on $\mathbb{R}$ no |
| $\sin(nx+n)/n$ | 0 on $\mathbb{R}$ | yes, on all of $\mathbb{R}$ |

# P1 Limits of sequences

> Every pointwise limit in this module is an ordinary sequence limit.

**REMIND ME · REAL ANALYSIS I, BARTLE 3.1**

$x_n \to x$ means: for every $\varepsilon > 0$ there is a natural number $K$ such that $|x_n - x| < \varepsilon$ for all $n \geq K$.

A sequence has at most **one** limit.

**STANDARD LIMITS TO KNOW BY HEART**

| | |
|---|---|
| $1/n \to 0$ | and $c/n \to 0$ for any fixed number $c$ |
| $b^n \to 0$ | when $\lvert b\rvert < 1$; $b^n = 1$ when $b = 1$ |
| $b^n$ | unbounded when $\lvert b\rvert > 1$; $(-1)^n$ has no limit |
| $n^{1/n} \to 1$ | $(1 + \frac{1}{n})^n \to e$ |
| $n\,e^{-na} \to 0$ | for each fixed $a > 0$ (exponentials beat powers) |

**CHECK YOURSELF**

**Find $\lim (0.9)^n$ and $\lim (1.1)^n$.**

$0.9^n \to 0$ ($|0.9| < 1$). $1.1^n$ grows without bound: no limit.

# P2 Algebra of limits and the squeeze

The two tools that do most pointwise limits.

**REMIND ME · BARTLE 3.2.3, 3.2.7**

- If $a_n \to a$ and $b_n \to b$ then $a_n + b_n \to a + b$, $a_n b_n \to ab$, and $a_n / b_n \to a/b$ when $b \neq 0$.
- **Squeeze**: if $|a_n| \leq b_n$ and $b_n \to 0$, then $a_n \to 0$.
- **Order limit** (3.2.6): if $a_n \leq c$ for all $n$ and $a_n \to a$, then $a \leq c$. Limits keep "≤" (but not "<").

**EXAMPLE**

$\dfrac{\cos(nx+n)}{n}$: since $|\cos| \leq 1$, we have $\left|\dfrac{\cos(nx+n)}{n}\right| \leq \dfrac{1}{n} \to 0$. By the squeeze the limit is 0, for every x.

**CHECK YOURSELF**

**Find $\lim \dfrac{nx}{1+nx}$ for a fixed $x > 0$.**

Divide top and bottom by $nx$: $\dfrac{1}{1/(nx)+1} \to \dfrac{1}{0+1} = 1$.

## P3 Supremum (least upper bound)

> The uniform norm is a supremum, so we need this exactly.

**REMIND ME · BARTLE 2.3**

- $u = \sup S$ means: $u$ is an upper bound of S ($s \le u$ for all $s \in S$), and no smaller number is an upper bound.
- So: $\sup S \le c \iff$ every $s \in S$ has $s \le c$.
- A sup need **not** be attained: $\sup\{x^n : 0 \le x < 1\} = 1$, but no $x$ in $[0, 1)$ gives $x^n = 1$.

**EXAM TRAP**

"$x^n < 1$ for every $x < 1$, so the sup is less than 1." **Wrong.** Each value is below 1, but they come as close to 1 as we like, so the sup is exactly 1.

**CHECK YOURSELF**

**Find $\sup\{x/(x+5) : x \ge 0\}$. Is it attained?**

$x/(x+5) = 1 - 5/(x+5)$ increases towards 1 as $x \to \infty$. Sup = 1, never attained.

# P4 Cauchy sequences

Used in the Cauchy criterion (T10).

**REMIND ME · BARTLE 3.5.1, 3.5.5**

$(a_n)$ is **Cauchy** if for every $\varepsilon > 0$ there is $H$ such that $|a_m - a_n| < \varepsilon$ for all $m, n \geq H$: late terms are close **to each other**.

**Cauchy Convergence Criterion (3.5.5)**: a sequence of real numbers converges if and only if it is Cauchy.

**THE IDEA**

With a Cauchy sequence you can prove convergence **without knowing the limit**. $\mathbb{R}$ has no holes, so a sequence whose terms bunch together must bunch round some real number.

# P5 Continuity at a point (from Module 1)

> Needed for T11–T14.

**REMIND ME · BARTLE 5.1.1**

$f$ is continuous at $c \in A$ if for every $\varepsilon > 0$ there is $\delta > 0$ such that $|f(x) - f(c)| < \varepsilon$ whenever $x \in A$ and $|x - c| < \delta$.

Polynomials, $\sin$, $\cos$, $e^x$, $\arctan$ and their sums, products and (non-zero denominator) quotients are continuous.

## P6 Integrals you can compute

> Only for T12 (two integral examples).

**REMIND ME · PLUS TWO, BARTLE 7.3.1**

- If $H' = h$ on $[a, b]$ then $\int_a^b h = H(b) - H(a)$ (Fundamental Theorem).
- $\dfrac{d}{dx}\left(-e^{-nx^2}\right) = 2nx\, e^{-nx^2}$ (chain rule).
- Area of a triangle = ½ × base × height.

# M1 Inequality kit: $|\cdot|$, ε/2 and ε/3

Three moves that appear in every proof of this module.

**MATHS TOOL**

- $|a| < \varepsilon$ means $-\varepsilon < a < \varepsilon$: "a is within ε of 0".
- **Triangle inequality**: $|a + b| \leq |a| + |b|$.
- **Add and subtract**: to compare $p$ and $r$ through a middle point $q$, write $p - r = (p - q) + (q - r)$, so $|p - r| \leq |p - q| + |q - r|$.
- **Split ε**: if each piece is less than $\varepsilon/2$, two pieces add to less than ε. With three pieces use $\varepsilon/3$.

**EXAMPLE**

If $|f_m(x) - f(x)| < \varepsilon/2$ and $|f_n(x) - f(x)| < \varepsilon/2$, then $|f_m(x) - f_n(x)| \leq |f_m(x) - f(x)| + |f(x) - f_n(x)| < \varepsilon$.

# M2 A function is one object; freeze x first

How to read $f_n(x)$ with two letters moving.

**MATHS TOOL**

$f_n(x)$ has two inputs: the frame number n and the point x.

- **Freeze x, let n run** → a sequence of numbers $f_1(x), f_2(x), \ldots$ (P1). Its limit is $f(x)$.
- **Freeze n, let x run** → one whole function $f_n$, a graph.

Pointwise convergence uses the first view. Uniform convergence compares whole graphs, the second view.

**EXAMPLE**

$f_n(x) = x/n$. Freeze $x = 6$: the numbers $6, 3, 2, 1.5, \ldots \to 0$. Freeze $n = 3$: the line $y = x/3$.

**EXAM TRAP**

Always say **which set** you work on. A is where the $f_n$ are defined, and the limit may exist only on a smaller set $A_0$. "Uniform" is always "uniform **on** some set".

# M3 The order of quantifiers (a game)

> The whole difference between pointwise and uniform is the order of three words.

**MATHS TOOL · THE GAME**

An opponent names a small $\varepsilon > 0$. You must answer with a stage K after which $|f_n(x) - f(x)| < \varepsilon$.

| CONVERGENCE | ORDER OF PLAY | K MAY DEPEND ON |
|---|---|---|
| **pointwise** | opponent picks ε **and** x, then you pick K | **ε and x** |
| **uniform** | opponent picks ε, you pick K, **then** opponent picks any x | **ε only** |

In symbols: pointwise is $\forall\varepsilon\ \forall x\ \exists K$; uniform is $\forall\varepsilon\ \exists K\ \forall x$. Only "for all x" has moved.

**EXAMPLE**

For $x/n$ on $\mathbb{R}$: given ε and x, $K > |x|/\varepsilon$ works. As $|x|$ grows, K must grow: no single K serves every x. Pointwise yes, uniform no.

# M4 Finding a supremum with calculus

How to compute $\|f_n - f\|$ in T7.

**MATHS TOOL · FOUR STEPS**

1. Write the gap $d_n(x) = |f_n(x) - f(x)|$ on the set.
2. Differentiate (in x, with n fixed) and solve $d_n'(x) = 0$: the critical points.
3. List the candidates: values at the critical points, at the end points, and the **limits** at open ends or at ∞.
4. The largest candidate is the sup. It may be a limit that is never reached.

**EXAMPLE**

$d_n(x) = x\,e^{-nx}$ on $[0, \infty)$. $d_n'(x) = e^{-nx}(1 - nx) = 0$ at $x = 1/n$. Candidates: $d_n(0) = 0$, $d_n(1/n) = \dfrac{1}{en}$, limit at ∞ = 0. So $\sup = \dfrac{1}{en}$.

**SHORT CUT · AM–GM**

$1 + n^2x^2 \geq 2nx$ (because $(1 - nx)^2 \geq 0$), so $\dfrac{nx}{1 + n^2x^2} \leq \dfrac{1}{2}$, with equality at $x = 1/n$.

## M5 Negating a statement

> To prove "not uniform" you must know exactly what "not uniform" says.

**MATHS TOOL · FLIP EACH WORD**

"for all" ↔ "there is", and the final inequality flips ($<$ becomes $\geq$).

Uniform: for every ε there is K such that for every $n \geq K$ and every x, $|f_n(x) - f(x)| < \varepsilon$.

**Not** uniform: there is an $\varepsilon_0$ such that for every K there are some $n \geq K$ and some x with $|f_n(x) - f(x)| \geq \varepsilon_0$.

**EXAM TRAP**

$\varepsilon_0$ is **one fixed number** (like ½). It must not shrink as n grows.

# Watching one point at a time

## T1 Pointwise convergence

The first definition (PYQ MTS-22 Q8, MAT-21 Q21).

**THE IDEA**

Stand at one point x. Watch the heights $f_1(x), f_2(x), f_3(x), \ldots$ If they settle to a number, call it $f(x)$. Do this at every point of a set $A_0$: the numbers $f(x)$ make a new function f, the **limit function**.

**DEFINITION 8.1.1 · SAY IT WORD FOR WORD**

Let $(f_n)$ be a sequence of functions on $A \subseteq \mathbb{R}$ to $\mathbb{R}$, let $A_0 \subseteq A$, and let $f : A_0 \to \mathbb{R}$. We say that the sequence $(f_n)$ **converges on** $A_0$ **to f** if, for each $x \in A_0$, the sequence $(f_n(x))$ converges to $f(x)$ in $\mathbb{R}$. Then f is called the **limit on** $A_0$ of $(f_n)$, and $(f_n)$ is said to **converge pointwise** on $A_0$.

In plain words: at every single point, the numbers $f_n(x)$ tend to $f(x)$.

**LEMMA 8.1.3 · THE E–K FORM**

$(f_n)$ converges on $A_0$ to f if and only if for each $\varepsilon > 0$ and each $x \in A_0$ there is a natural number $K(\varepsilon, x)$ such that if $n \geq K(\varepsilon, x)$, then $|f_n(x) - f(x)| < \varepsilon$.

Note the label $K(\varepsilon, x)$: the stage K is allowed to depend on **x** as well as ε (M3).

**WORKED EXAMPLE · FINDING K(E, X)**

$f_n(x) = x/n$, $f = 0$ on $\mathbb{R}$. We need $|x/n - 0| < \varepsilon$, i.e. $n > |x|/\varepsilon$.

So $K(\varepsilon, x)$ = any whole number bigger than $|x|/\varepsilon$ works. At $x = 1$, $\varepsilon = 0.1$: $K = 11$. At $x = 100$: $K = 1001$.

The further out x is, the longer we wait. No single K works for all x: we will see this means "not uniform" (T3).

**RESULT · UNIQUENESS**

A sequence of numbers has at most one limit (P1). So the limit function f, if it exists, is unique.

**CHECK YOURSELF**

**For $f_n(x) = x/n$ and $\varepsilon = 0.01$, find a K that works at x = 3.**

Need $n > 3/0.01 = 300$. K = 301.

In the model paper: **Q2** (PYQ MTS-22 Q8), **Q13**.

## T2 Finding the limit function

> The most frequent past question in this module (asked six times).

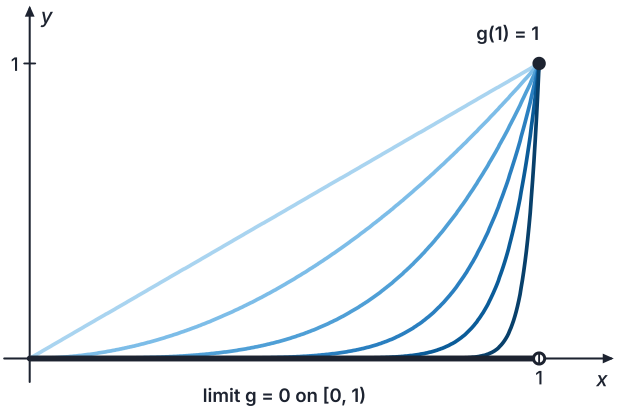
**SKELETON · FIND THE POINTWISE LIMIT**

1. Fix $x \in A$. (Now $f_n(x)$ is a sequence of numbers.)
2. Split into cases where the behaviour changes (often $x = 0$, $x = 1$, $x = -1$, $|x| < 1$, $|x| > 1$).
3. In each case compute $\lim_n f_n(x)$ with P1 and P2.
4. Write f piecewise, and state $A_0$, the set where the limit exists.

**WORKED EXAMPLE · BARTLE 8.1.2(B): $g_n(x) = x^n$ ON $\mathbb{R}$**

| | |
|---|---|
| $x = 1$ | $1^n = 1 \to 1$ |
| $\|x\| < 1$ | $x^n \to 0$ (P1) |
| $x = -1$ | $(-1)^n$ = −1, 1, −1, ...: no limit |
| $\|x\| > 1$ | unbounded: no limit |

So $A_0 = (-1, 1]$, with $g(x) = 0$ for $-1 < x < 1$ and $g(1) = 1$.




$g_n(x) = x^n$ on [0, 1] for n = 1, 2, 4, 8, 16, 40 (darker = larger n). For each fixed x < 1 the values fall to 0, but every curve still climbs to (1, 1). The limit g (black) has a **jump** at 1, although every $g_n$ is continuous C12.

## Three more, the way past papers ask them

$\lim e^{-nx}$, $x \geq 0$ **(PYQ MTS-25 Q10)**

At $x = 0$: $e^0 = 1$, limit 1. For $x > 0$: $e^{-nx} = (e^{-x})^n$ with $0 < e^{-x} < 1$, so the limit is 0. Limit: $f(0) = 1$, $f(x) = 0$ for $x > 0$.

$\lim \frac{nx}{1+nx}$, $x \geq 0$ **(PYQ MTS-26 Q10)**

At $x = 0$: every term is 0. For $x > 0$: $\frac{1}{1 + 1/(nx)} \to 1$. Limit: $f(0) = 0$, $f(x) = 1$ for $x > 0$.

$\lim \frac{x^2+nx}{n}$ **(PYQ MAT-22 Q8; BARTLE 8.1.2(C))**

$\frac{x^2+nx}{n} = \frac{x^2}{n} + x \to 0 + x = x$ for every $x \in \mathbb{R}$.

**EXAM TRAP**

The exceptional points ($x = 0$ above) are where marks are lost. Before writing "the limit is 0", test the end points and the points where a base equals 1.

**CHECK YOURSELF**

**Find $\lim(\cos \pi x)^{2n}$ for $x \in \mathbb{R}$.**

Put $t = \cos^2 \pi x \in [0, 1]$, so the term is $t^n$. If x is an integer, $t = 1$ and the limit is 1; otherwise $0 \leq t < 1$ and the limit is 0.

In the model paper: **Q4** **Q5** **Q6** **Q7** **Q16** **Q21** .

# One K for every point

## T3 Uniform convergence

> The central definition of the module (PYQ MTS-26 Q9).

**DEFINITION 8.1.4 · SAY IT WORD FOR WORD**

A sequence $(f_n)$ of functions on $A \subseteq \mathbb{R}$ to $\mathbb{R}$ **converges uniformly on** $A_0 \subseteq A$ to a function $f : A_0 \to \mathbb{R}$ if for each $\varepsilon > 0$ there is a natural number $K(\varepsilon)$ (depending on ε but not on $x \in A_0$) such that if $n \geq K(\varepsilon)$, then $|f_n(x) - f(x)| < \varepsilon$ for all $x \in A_0$.

In plain words: one stage K, chosen from ε alone, after which **every** point x is within ε. Picture: the whole graph of $f_n$ sits in the ε-tube round f.

**SKELETON · PROVE $f_n \to f$ UNIFORMLY ON $A_0$**

Let $\varepsilon > 0$. Choose $K =$ ? (built from ε only). Let $n \geq K$ and $x \in A_0$. Then $|f_n(x) - f(x)| \leq$ ? $< \varepsilon$. Since K does not depend on x, $f_n \to f$ uniformly on $A_0$. ∎

**COMPARE THE TWO DEFINITIONS**

| | K DEPENDS ON | PICTURE |
|---|---|---|
| **pointwise** | **ε and x** | each point settles, in its own time |
| **uniform** | **ε only** | whole graph inside the ε-tube |

**RESULT · UNIFORM ⇒ POINTWISE**

If $f_n \to f$ uniformly on $A_0$, then $f_n \to f$ pointwise on $A_0$ (the same K works at each x). The converse is **false**: $x/n$ on $\mathbb{R}$, $x^n$ on $[0,1)$.

**EXAM TRAP**

Writing $K = |x|/\varepsilon$ and calling it uniform. Before writing "uniform", **circle every letter in your K**: only ε and fixed numbers are allowed.

**CHECK YOURSELF**

**Show $\frac{1}{n}\sin(nx+n) \to 0$ uniformly on $\mathbb{R}$.**

Let $\varepsilon > 0$; take $K > 1/\varepsilon$. For $n \geq K$ and every x: $|\frac{1}{n}\sin(nx+n)| \leq \frac{1}{n} \leq \frac{1}{K} < \varepsilon$. K uses ε only. ∎

In the model paper: **Q3** (PYQ MTS-26 Q9).

# T4 Proving "NOT uniform"

> Find points where $f_n$ stays far away (Lemma 8.1.5; PYQ MTS-22 Q9).

**LEMMA 8.1.5 · SAY IT WORD FOR WORD**

A sequence $(f_n)$ of functions on $A \subseteq \mathbb{R}$ to $\mathbb{R}$ does **not** converge uniformly on $A_0 \subseteq A$ to a function $f : A_0 \to \mathbb{R}$ if and only if for some $\varepsilon_0 > 0$ there is a subsequence $(f_{n_k})$ of $(f_n)$ and a sequence $(x_k)$ in $A_0$ such that $|f_{n_k}(x_k) - f(x_k)| \geq \varepsilon_0$ for all $k \in \mathbb{N}$.

In plain words: a fixed distance $\varepsilon_0$, and for frames as late as you like, a point $x_k$ where the frame is still at least $\varepsilon_0$ away.

**SKELETON · PROVE NOT UNIFORM ON $A_0$**

Take $\varepsilon_0 =$ ? , $n_k = k$ and $x_k =$ ? $\in A_0$. Then $|f_k(x_k) - f(x_k)| =$ ? $\geq \varepsilon_0$ for all k. By Lemma 8.1.5, $(f_n)$ does not converge uniformly on $A_0$. ∎

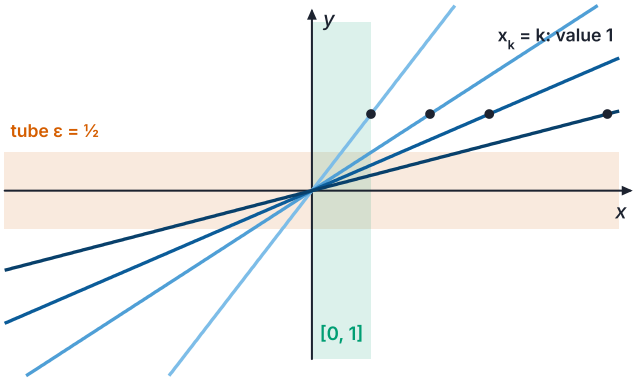
**SCRATCH · HOW TO CHOOSE $x_k$**

Look for where $f_k$ is still big: at its peak, far out towards ∞, or near the bad end point. Good recipes: solve $f_k(x) = \frac{1}{2}$; or try $x_k = k$, $x_k = 1/k$, $x_k = -k$.

## Bartle's three examples (8.1.6)

**(A) $x/n$ ON $\mathbb{R}$, LIMIT 0**

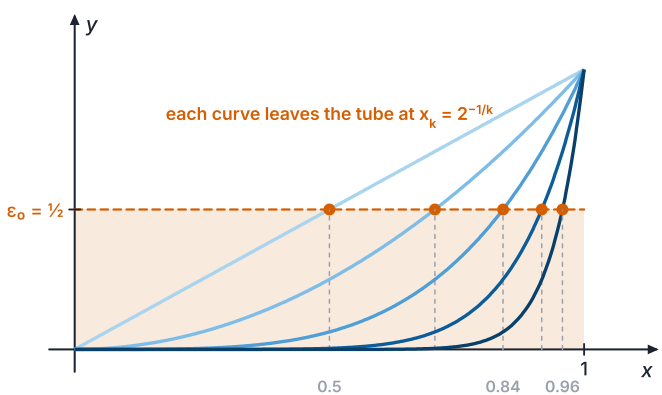
$x_k = k$: $|f_k(k) - 0| = k/k = 1$. Take $\varepsilon_0 = 1$. Not uniform on $\mathbb{R}$.




x/n → 0 at every x, but the line y = x/n always escapes the ε-tube far out (at x = k the value is 1). Not uniform on ℝ. On [0, 1] (green) the worst gap is 1/n → 0: uniform there **C15**.

**(B) $x^n$ ON (−1, 1], LIMIT G**

$x_k = (1/2)^{1/k}$, so $x_k^k = \frac{1}{2}$ and $g(x_k) = 0$ (since $x_k < 1$). Gap $\frac{1}{2} = \varepsilon_0$. Not uniform.




Why $x^n$ is not uniform on [0, 1): with $\varepsilon_o$ = ½ the tube is 0 ≤ y < ½. Each $x^k$ crosses ½ at $x_k = 2^{-1/k}$ (0.5, 0.71, 0.84, 0.92, 0.96 ...). The escape point moves towards 1 but never disappears, so no K works for all x.

**(C) $(x^2+nx)/n$ ON $\mathbb{R}$, LIMIT X**

$x_k = -k$: $h_k(-k) = (k^2 - k^2)/k = 0$ and $h(-k) = -k$, so the gap is $k \geq 1 = \varepsilon_0$. Not uniform on $\mathbb{R}$.

**PROOF · THE LEMMA ITSELF (5 MARKS)**

1. "Uniform" says: for all ε there is K with $|f_n(x) - f(x)| < \varepsilon$ for all $n \geq K$, all $x \in A_0$.
   Definition 8.1.4.

2. Its negation: there is $\varepsilon_0 > 0$ such that for every K some $n \geq K$ and some $x \in A_0$ have $|f_n(x) - f(x)| \geq \varepsilon_0$.
   Flip each quantifier and the inequality (M5).

3. K = 1 gives $n_1$, $x_1$. K = $n_1 + 1$ gives $n_2 > n_1$, $x_2$. Continue.
   Builds an increasing $n_k$: a subsequence.

4. Conversely, given such $n_k, x_k$: for any K pick k with $n_k \geq K$; that pair breaks uniformity.
   $n_k \to \infty$, so late enough frames always exist.

**EXAM TRAP**

$\varepsilon_0$ must be **one fixed number**. "Gap $= 1/k$" proves nothing; "gap $= \frac{1}{2}$ for every k" proves it.

In the model paper: **Q9** (PYQ MTS-22 Q9), **Q24**, **Q26**, **Q34**.

# T5 Bounded functions and the uniform norm

One number that measures the worst gap (PYQ MTS-25 Q9).

**DEFINITION 8.1.7 · SAY IT WORD FOR WORD**

If $A \subseteq \mathbb{R}$ and $\varphi : A \to \mathbb{R}$ is a function, we say that φ is **bounded on A** if the set $\varphi(A)$ is a bounded subset of $\mathbb{R}$. If φ is bounded we define the **uniform norm of φ on A** by

$$\|\varphi\|_A := \sup\{|\varphi(x)| : x \in A\}.$$

In plain words: the height of the highest point of the graph of $|\varphi|$.

**RESULT · THE KEY PROPERTY**

$$\|\varphi\|_A \leq \varepsilon \iff |\varphi(x)| \leq \varepsilon \text{ for all } x \in A.$$

(A sup is ≤ ε exactly when every element is ≤ ε, P3.)

**CHECK YOURSELF**

**Find $\|\sin\|_{\mathbb{R}}$, $\|x\|_{[-2,1]}$ and $\|x^n\|_{[0,1)}$.**

1 (attained at π/2). 2 (at x = −2). 1 (sup, not attained).

In the model paper: **Q1** (PYQ MTS-25 Q9).

# T6 The norm test (Lemma 8.1.8)

> Uniform convergence ⇔ worst gap → 0 (PYQ MTS-26 Q20, MAT-22 Q19).

**LEMMA 8.1.8 · SAY IT WORD FOR WORD**

A sequence $(f_n)$ of bounded functions on $A \subseteq \mathbb{R}$ converges uniformly on A to f if and only if $\|f_n - f\|_A \to 0$.

**Know** $f_n$ bounded on A; T3 definition; T5 key property.

**Want** uniform ⇔ the real sequence $\|f_n - f\|_A$ tends to 0.

**PROOF (⇒)**

1. Let $\varepsilon > 0$. By uniformity there is $K(\varepsilon)$ with $|f_n(x) - f(x)| \le \varepsilon$ for all $x \in A$, $n \ge K(\varepsilon)$.
   Definition 8.1.4 (we only need ≤).

2. So ε is an upper bound of $\{|f_n(x) - f(x)| : x \in A\}$, hence $\|f_n - f\|_A \le \varepsilon$ for $n \ge K(\varepsilon)$.
   The sup is the **least** upper bound (P3).

3. ε was arbitrary, so $\|f_n - f\|_A \to 0$.
   This is the ε–K definition of a limit of numbers (P1).

**PROOF (⇐)**

1. Let $\varepsilon > 0$. Since $\|f_n - f\|_A \to 0$, there is $H(\varepsilon)$ with $\|f_n - f\|_A \leq \varepsilon$ for $n \geq H(\varepsilon)$.
   Definition of limit (P1).

2. For every $x \in A$: $|f_n(x) - f(x)| \leq \|f_n - f\|_A \leq \varepsilon$ when $n \geq H(\varepsilon)$.
   Each element of a set is ≤ its sup.

3. H depends on ε only, so $f_n \to f$ uniformly on A. ∎
   Matches Definition 8.1.4 (use ε/2 if you want strict <).

**HYPOTHESES CHECKLIST**

☐ $f_n - f$ bounded on A
otherwise the "sup" is $+\infty$ and the norm is not a real number. Example: $x/n$ on $\mathbb{R}$: $f_n - f$ is unbounded, so automatically **not** uniform C15.

**WHY THIS LEMMA IS SO USEFUL**

It turns a question about functions (for all ε, all x ...) into a question about one sequence of numbers $\|f_n - f\|$, which you can compute with calculus (T7).

In the model paper: **Q39** (PYQ MTS-26 Q20, also MAT-22 Q19).

## T7 Computing $\|f_n - f\|$

The standard 6–9 mark problem (PYQ MAT-21 Q32).

**SKELETON · "DISCUSS THE CONVERGENCE OF $f_n$ ON A"**

1. **Pointwise**: fix x, find f (T2).
2. **Gap**: write $d_n(x) = |f_n(x) - f(x)|$.
3. **Worst gap**: $\|f_n - f\|_A = \sup d_n$ by calculus (M4).
4. **Verdict**: if it $\to 0$, uniform (Lemma 8.1.8); if not, not uniform.

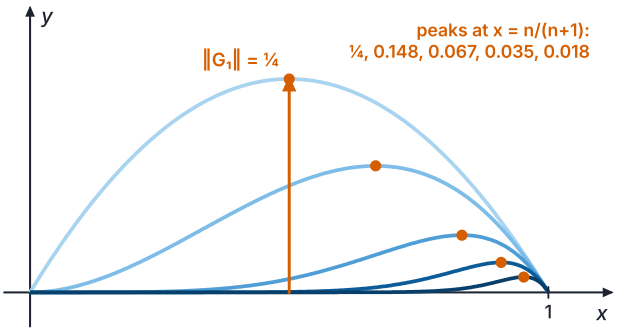
**WORKED EXAMPLE · BARTLE 8.1.9(E): $G_n(x) = x^n(1-x)$ ON [0, 1]**

**1.** At $x = 0, 1$: $G_n = 0$. For $0 < x < 1$: $x^n \to 0$. So $G = 0$.

**2–3.** $G_n'(x) = x^{n-1}\big(n - (n+1)x\big) = 0$ at $x = \dfrac{n}{n+1}$, a maximum.

$$\|G_n\| = \left(\frac{n}{n+1}\right)^n \frac{1}{n+1} = \frac{n^n}{(n+1)^{n+1}}.$$

**4.** $\left(\dfrac{n}{n+1}\right)^n \le 1$, so $\|G_n\| \le \dfrac{1}{n+1} \to 0$: **uniform** on [0, 1].

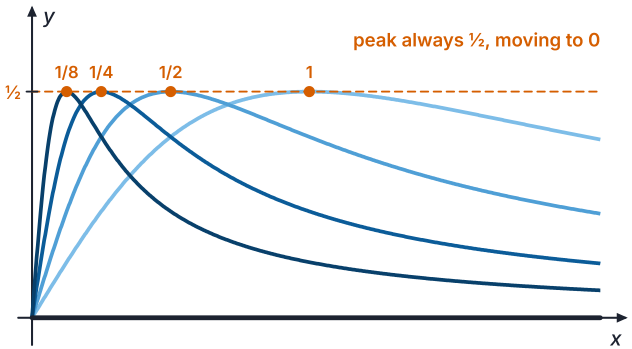



The uniform norm is the **height of the highest point** of $|f_n - f|$. For $G_n = x^n(1 - x)$ the peak sits at $x = n/(n+1)$ with height $n^n/(n+1)^{n+1} \le 1/(n+1) \to 0$, so $G_n \to 0$ uniformly on $[0, 1]$.

**WORKED EXAMPLE · THE MOVING BUMP $f_n(x) = \frac{nx}{1 + n^2x^2}$ ON $[0, \infty)$**

Pointwise: $f_n(0) = 0$; for $x > 0$, $0 \le f_n(x) \le \frac{nx}{n^2x^2} = \frac{1}{nx} \to 0$. So $f = 0$.

Worst gap: by AM–GM (M4), $f_n \le \frac{1}{2}$ with equality at $x = 1/n$. So $\|f_n\| = \frac{1}{2}$ for every n: **not uniform** on $[0, \infty)$.




The moving bump $f_n(x) = nx/(1 + n^2x^2)$ on $[0, \infty)$. For each fixed $x > 0$, $f_n(x) \to 0$. But every $f_n$ has a peak of height ½ at $x = 1/n$, so $\|f_n - 0\| = ½$ for all n: **pointwise, not uniform** C21.

**EXAM TRAP**

A sup can be a limit that is never reached: $\sup\{x/(x+n) : x \geq 0\} = 1$. Check the ends and ∞, not just the critical points.

**CHECK YOURSELF**

**Find $\|xe^{-nx}\|_{[0,\infty)}$. Uniform to 0?**

Max at $x = 1/n$: value $\dfrac{1}{en} \to 0$. Uniform.

In the model paper: Q14 Q18 Q22 (PYQ MAT-21 Q32) Q26 Q33.

# T8 The shortcut: a bound free of x

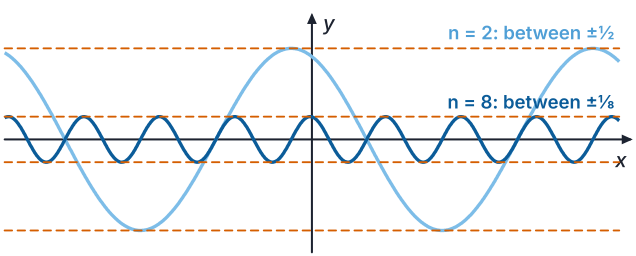
> When you can bound the gap by something with no x in it, you are done.

**RESULT**

If $|f_n(x) - f(x)| \le a_n$ for all $x \in A$, and $a_n \to 0$, then $\|f_n - f\|_A \le a_n \to 0$, so $f_n \to f$ **uniformly** on A.

**WORKED EXAMPLE · BARTLE 8.1.9(D)**

$F_n(x) = \frac{1}{n}\sin(nx + n)$: $|F_n(x)| \le \frac{1}{n}$ for every real x. So $\|F_n\|_{\mathbb{R}} \le \frac{1}{n} \to 0$: uniform on $\mathbb{R}$. (In fact $\|F_n\| = \frac{1}{n}$, reached when $nx + n = \pi/2$.)




$F_n(x) = \sin(nx + n)/n$ lies between the lines $y = \pm 1/n$ (dashed) for **every** x at once. The bound 1/n does not depend on x, so $\|F_n\| = 1/n \to 0$: uniform on $\mathbb{R}$.

**EXAM TRAP**

The bound must not contain x. $|x/n| \le |x|/n$ is useless on $\mathbb{R}$ (it grows with x), but on [0, 1] it gives $\le 1/n$.

In the model paper: **Q8** (PYQ MTS-22 MCQ 12, also MAT-21 Q10), **Q32** .

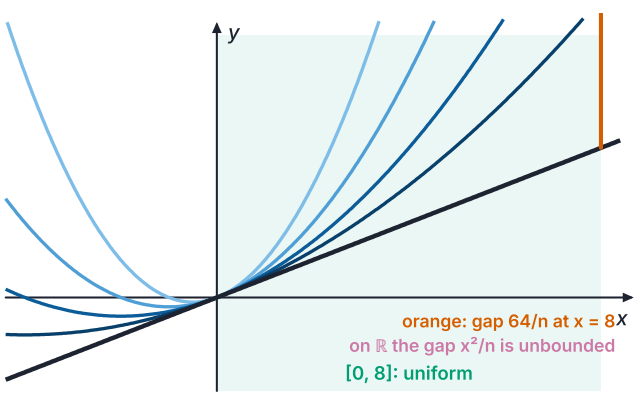
# T9 Uniform depends on the set

> The same sequence can be uniform on one set and not on another (PYQ MAT-21 Q37b).

**THE IDEA**

Uniformity fails because of a **bad place**: far out towards ∞, or near a point where the limit jumps. Cut the bad place out of the set and uniformity comes back.

| $F_N$ | UNIFORM ON | NOT UNIFORM ON |
|---|---|---|
| $x/n$ | [0, 1]: norm $1/n$; any $[-b, b]$: $b/n$ | $\mathbb{R}$ |
| $x^n$ | $[0, b]$, $b < 1$: norm $b^n$ | [0, 1], and even [0, 1) |
| $(x^2 + nx)/n$ | [0, 8]: norm $64/n$ | $\mathbb{R}$ |
| $nx/(1 + n^2x^2)$ | $[a, \infty)$, $a > 0$ | $[0, \infty)$ |




$h_n(x) = (x^2 + nx)/n = x + x^2/n \to x$ (black). The gap $h_n - h = x^2/n$ is at most $64/n$ on [0, 8], so uniform there; on $\mathbb{R}$ it has no upper bound, so not uniform.

**WORKED EXAMPLE · $x^n$ ON THREE SETS**

**On [0, b], b < 1.** $x^n$ increases in x, so $\|x^n - 0\|_{[0,b]} = b^n \to 0$. Uniform. (For $b = \frac{1}{2}$: norm $2^{-n}$.)

**On [0, 1).** The limit is 0 (continuous!), but $\sup\{x^n : 0 \le x < 1\} = 1$ for every n. Not uniform C25.

**On [0, 1].** Gap $|x^n - g(x)|$ is $x^n$ on [0, 1) and 0 at 1, so the norm is 1. Not uniform.

**EXAM TRAP**

"The limit is continuous, so the convergence is uniform." False: $x^n$ on [0, 1) has the continuous limit 0 and is still not uniform.

**CHECK YOURSELF**

**Is $x/n \to 0$ uniform on [−5, 5]?**

Yes: $|x/n| \le 5/n$ for all $x \in [-5, 5]$, so the norm is $5/n \to 0$.

In the model paper: Q17 Q19 Q28 Q40 (PYQ MAT-21 Q37b).

# T10 The Cauchy criterion (Theorem 8.1.10)

> A 10–15 mark favourite (PYQ MTS-25 Q26, MAT-21 Q38a).

**THEOREM 8.1.10 · SAY IT WORD FOR WORD**

Let $(f_n)$ be a sequence of bounded functions on $A \subseteq \mathbb{R}$. Then this sequence converges uniformly on A to a bounded function f if and only if for each $\varepsilon > 0$ there is a number $H(\varepsilon)$ in $\mathbb{N}$ such that for all $m, n \geq H(\varepsilon)$, then $\|f_m - f_n\|_A \leq \varepsilon$.

In plain words: late frames are uniformly close **to each other**. Like P4, but for whole graphs. You never need to know f.

**Know** Lemma 8.1.8 (T6); Cauchy Criterion 3.5.5 for numbers (P4); order limit 3.2.6 (P2); triangle inequality (M1).

**Want** (⇒) uniform ⇒ uniformly Cauchy. (⇐) uniformly Cauchy ⇒ there is a bounded f with $f_n \to f$ uniformly.

**PROOF (⇒)**

1. Let $\varepsilon > 0$. By Lemma 8.1.8 there is $K = K(\varepsilon/2)$ with $\|f_n - f\|_A \le \varepsilon/2$ for $n \ge K$.
   Use ε/2 so two pieces add to ε (M1).

2. For $m, n \ge K$ and every $x \in A$: $|f_m(x) - f_n(x)| \le |f_m(x) - f(x)| + |f(x) - f_n(x)| \le \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.
   Add and subtract f(x); triangle inequality.

3. Taking sup over x: $\|f_m - f_n\|_A \le \varepsilon$ for $m, n \ge H(\varepsilon) := K(\varepsilon/2)$.
   ε bounds every value, so it bounds the sup (T5).

**PROOF (⇐)**

1. Fix $x \in A$. For $m, n \geq H(\varepsilon)$: $|f_m(x) - f_n(x)| \leq \|f_m - f_n\|_A \leq \varepsilon$. So $(f_n(x))$ is a Cauchy sequence of numbers.
   Each value ≤ the sup; definition of Cauchy (P4).

2. By the Cauchy Convergence Criterion 3.5.5 it converges. Define $f(x) := \lim_n f_n(x)$ for each $x \in A$.
   $\mathbb{R}$ is complete. This is our candidate f.

3. In $|f_m(x) - f_n(x)| \leq \varepsilon$ (m fixed, $m \geq H$) let $n \to \infty$: $|f_m(x) - f(x)| \leq \varepsilon$ for all $x \in A$, $m \geq H(\varepsilon)$.
   Order limit theorem 3.2.6: limits keep ≤. **The same H works for every x**: this is the heart of the proof.

4. f is bounded: with $m = H(1)$, $|f(x)| \leq |f_m(x)| + 1 \leq \|f_m\|_A + 1$ for all x.
   Step 3 with ε = 1; $f_m$ is bounded. (Bartle leaves this line out; write it.)

5. So $\|f_m - f\|_A \leq \varepsilon$ for $m \geq H(\varepsilon)$, i.e. $f_n \to f$ uniformly. ∎
   Lemma 8.1.8.

**EXAM TRAP**

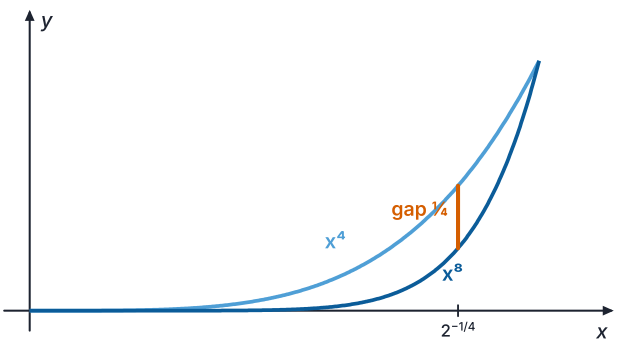
"$\|f_{n+1} - f_n\| \to 0$" is **not** the Cauchy condition. You need **all** pairs $m, n \geq H$. (Constant functions $f_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ have neighbours closer and closer, but they never converge.)

**USING THE CRITERION · BARTLE-STYLE**

Show $(x^n)$ is not uniformly convergent on [0, 1] **without** finding its limit. Put $t = x^n \in [0,1]$:

$$x^n - x^{2n} = t - t^2 \le \tfrac{1}{4},$$

with equality at $t = \dfrac{1}{2}$, i.e. $x = 2^{-1/n}$. So $\|f_n - f_{2n}\| = \dfrac{1}{4}$ for every n. With $\varepsilon = \dfrac{1}{8}$ no H works (take m = H, n = 2H). Not uniformly Cauchy, so not uniform.




Not uniformly Cauchy. $\|x^n - x^{2n}\|$ = max of $t - t^2$ ($t = x^n$) = ¼, reached where $x^n$ = ½. However large n is, $f_n$ and $f_{2n}$ stay ¼ apart somewhere, so the Cauchy criterion fails (with ε = ⅛, say).

In the model paper: **Q31** , **Q37** (PYQ MTS-25 Q26, also MAT-21 Q38a).

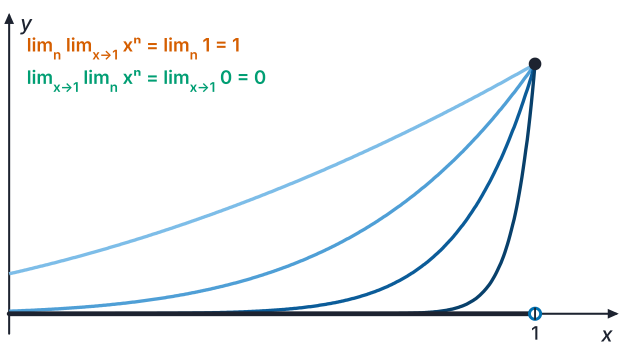
# What survives the limit?

## T11 Pointwise limits can break continuity

Bartle 8.2.1(a), (b) (PYQ MTS-22 MCQ 11).

**THE IDEA**

Every $g_n(x) = x^n$ is continuous (even smooth) on [0, 1]. Their pointwise limit g is 0 on [0, 1) and 1 at x = 1: it **jumps**. So a pointwise limit of continuous functions need not be continuous, and need not be differentiable.




The order of two limits matters. Take x → 1 first and you get 1 for every n; take n → ∞ first and you get 0 for every x < 1. Pointwise convergence lets these differ; this is why continuity is lost.

**RESULT · TWO LIMITS IN THE WRONG ORDER**

$$\lim_{x\to 1^-}\left(\lim_{n} x^n\right) = 0, \qquad \lim_{n}\left(\lim_{x\to 1^-} x^n\right) = 1.$$

"Continuity of the limit" is exactly the statement that these two orders agree.

**CHECK YOURSELF · PAST MCQ**

**$g_n(x) = x^n$. On which of (−1, 1], (−2, 1], (0, 1], (−1, 1) does $(g_n)$ converge to a continuous function?**

Only (−1, 1): there the limit is 0. On (−1, 1] and (0, 1] the limit jumps at 1; on (−2, 1] there is no limit at x ≤ −1.

**EXAM TRAP**

Derivatives are worse still: $\frac{1}{n}\sin nx \to 0$ uniformly on $\mathbb{R}$, but the derivatives $\cos nx$ do not converge (at $x = \pi$ they are $(-1)^n$). Uniform convergence of $f_n$ alone does not let you differentiate the limit term by term.

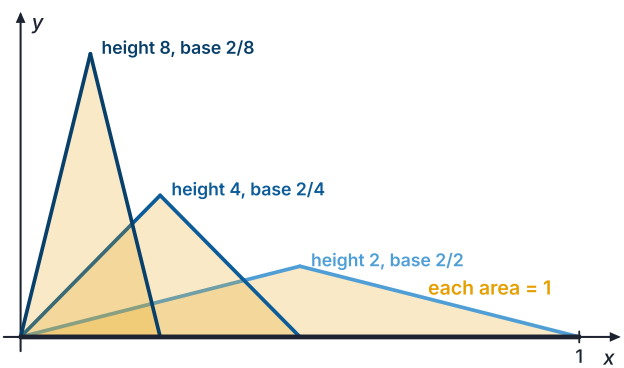
In the model paper: **Q11** (PYQ MTS-22 MCQ 11), **Q25**, **Q32**.

# T12 Limits and integrals

> $\lim \int f_n$ need not equal $\int \lim f_n$ (Bartle 8.2.1(c), (d); PYQ MTS-22 Q10).

**EXAMPLE 1 · SPIKES (8.2.1(C))**

$f_n$ is the triangle with corners (0, 0), (1/n, n), (2/n, 0), and 0 on [2/n, 1]. Area = ½ · (2/n) · n = 1 for every n. At a fixed $x > 0$, once $2/n < x$ the spike has passed and $f_n(x) = 0$; also $f_n(0) = 0$. Limit f = 0, $\int_0^1 f = 0 \neq 1$.



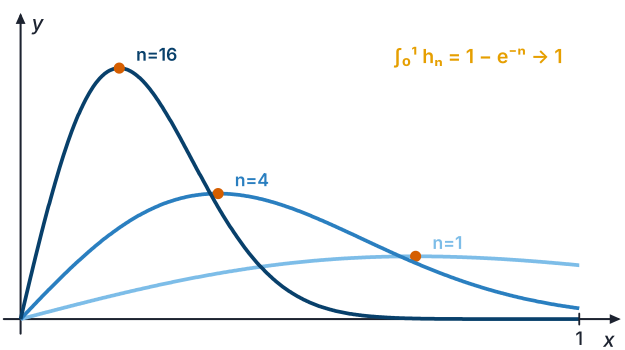

Spikes (Bartle 8.2.1(c)). Each $f_n$ is a triangle of base 2/n and height n, so $\int_0^1 f_n = 1$. At any fixed x > 0 the spike has passed once 2/n < x, so $f_n(x) \to 0$. $\lim \int f_n = 1$ but $\int \lim f_n = 0$ C13.

**EXAMPLE 2 · $h_n(x) = 2nx\,e^{-nx^2}$ ON [0, 1] (PYQ MTS-22 Q10)**

**Integral.** $H_n(x) = -e^{-nx^2}$ has $H_n' = h_n$ (P6), so $\int_0^1 h_n = H_n(1) - H_n(0) = 1 - e^{-n} \to 1$.

**Limit.** $h_n(0) = 0$. For $x > 0$: $e^{nx^2} > \dfrac{(nx^2)^2}{2}$ (one term of the exponential series), so $0 < h_n(x) < \dfrac{2nx \cdot 2}{n^2x^4} = \dfrac{4}{nx^3} \to 0$. So $h = 0$ and $\int_0^1 h = 0$.

$\lim \int h_n = 1 \neq 0 = \int \lim h_n$.




$h_n(x) = 2nx\,e^{-nx^2}$: peak $\sqrt{(2n/e)}$ at $x = 1/\sqrt{(2n)}$, moving to 0 and growing. $h_n(x) \to 0$ for every x, yet $\int_0^1 h_n = 1 - e^{-n} \to 1 \neq 0$.

**EXAM TRAP**

Uniform convergence is **enough** to swap limit and integral (Theorem 8.2.4, practicum), but it is not **needed**: $\dfrac{nx}{1+nx}$ on [0, 1] is not uniform, yet $\int_0^1 f_n = 1 - \dfrac{\ln(1+n)}{n} \to 1 = \int_0^1 f$ C26.

In the model paper: **Q10** (PYQ MTS-22 Q10), **Q27**, **Q35**.

# T13 Uniform limit of continuous functions is continuous

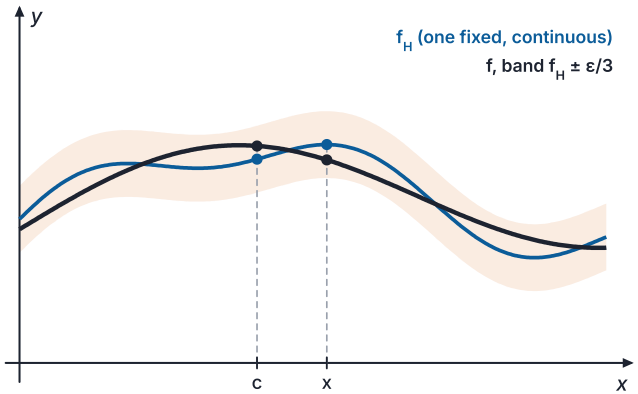
> Theorem 8.2.2, the main theorem of §8.2 (PYQ MTS-25 Q20).

**THEOREM 8.2.2 · SAY IT WORD FOR WORD**

Let $(f_n)$ be a sequence of continuous functions on a set $A \subseteq \mathbb{R}$ and suppose that $(f_n)$ converges uniformly on A to a function $f : A \to \mathbb{R}$. Then f is continuous on A.

**HYPOTHESES CHECKLIST**

- ☐ every $f_n$ continuous on A
  without it nothing is inherited: constant step functions converging to a step function.
- ☐ convergence is **uniform** on A
  pointwise is not enough: $x^n$ on [0, 1] has a jump limit C12.




Proof of Theorem 8.2.2: the ε/3 trick. Pick ONE $f_H$ within ε/3 of f **everywhere** (uniformity). Then $|f(x) - f(c)| \le |f(x) - f_H(x)| + |f_H(x) - f_H(c)| + |f_H(c) - f(c)|$: outer two < ε/3 by uniformity, middle < ε/3 by continuity of $f_H$.

**Know** uniform: one H with $|f_H(x) - f(x)| < \varepsilon/3$ for **all** x; $f_H$ continuous at c.

**Want** for each $c \in A$ and ε > 0, a δ with $|f(x) - f(c)| < \varepsilon$ when $|x - c| < \delta$, $x \in A$.

**PROOF (BARTLE'S E/3 ARGUMENT)**

1. Let $\varepsilon > 0$ and $c \in A$.
   To show continuity at an arbitrary c (P5).

2. By uniform convergence there is $H = H(\varepsilon/3)$ with $|f_H(x) - f(x)| < \varepsilon/3$ for **all** $x \in A$.
   Definition 8.1.4 with ε/3. Fix this one function $f_H$.

3. For $x \in A$: $|f(x) - f(c)| \leq |f(x) - f_H(x)| + |f_H(x) - f_H(c)| + |f_H(c) - f(c)| < \dfrac{\varepsilon}{3} + |f_H(x) - f_H(c)| + \dfrac{\varepsilon}{3}$.
   Add and subtract $f_H(x)$ and $f_H(c)$ (M1).
   Uniformity is used at **two** points, x and c.

4. $f_H$ is continuous at c, so there is $\delta > 0$ with $|f_H(x) - f_H(c)| < \varepsilon/3$ when $x \in A$, $|x - c| < \delta$.
   Continuity of the single function $f_H$.

5. Then $|f(x) - f(c)| < \dfrac{\varepsilon}{3} + \dfrac{\varepsilon}{3} + \dfrac{\varepsilon}{3} = \varepsilon$ for those x. c was arbitrary, so f is continuous on A. ∎
   Steps 3 and 4 together.

**WHERE POINTWISE WOULD FAIL**

In step 2 we need one H that works at x **and** at c, for every x near c. With only pointwise convergence, H would depend on x, and x moves as δ shrinks.

In the model paper: **Q38** (PYQ MTS-25 Q20).

# T14 Using Theorem 8.2.2

> The quickest proof of "not uniform" (PYQ MAT-22 Q31).

**SKELETON · CONTRAPOSITIVE**

Each $f_n$ is continuous on A (reason). The pointwise limit is f = ?, which is **not** continuous at c. If $f_n \to f$ uniformly on A, Theorem 8.2.2 would make f continuous. Hence the convergence is not uniform on A. ∎

**WORKED EXAMPLE · $f_n(x) = \frac{x^n}{x^n+1}$ ON [0, 2] (PYQ MAT-22 Q31)**

$0 \le x < 1$ $\quad x^n \to 0$, so $f_n(x) \to 0$

$x = 1$ $\quad f_n(1) = \frac{1}{2}$

$1 < x \le 2$ $\quad f_n(x) = \frac{1}{1+x^{-n}} \to 1$

The $f_n$ are continuous; the limit jumps at 1. So **not uniform** on [0, 2]. (Directly: $\|f_n - f\| = \frac{1}{2}$ for every n.) On $[0, b]$ with $b < 1$ the norm is $\le b^n \to 0$: uniform there.

**EXAM TRAP · THE CONVERSE IS FALSE**

"The limit is continuous, so the convergence is uniform." **No.** $x/n \to 0$ on $\mathbb{R}$: limit continuous, convergence not uniform. Theorem 8.2.2 can only **disprove** uniformity (when the limit is discontinuous); it can never prove it.

**CHECK YOURSELF**

**Show $\arctan(nx) \to \frac{\pi}{2}\operatorname{sgn} x$ and that the convergence is not uniform on $\mathbb{R}$.**

x > 0: $nx \to \infty$, $\arctan \to \pi/2$. x < 0: $-\pi/2$. x = 0: 0. The $f_n$ are continuous but the limit jumps at 0, so not uniform (8.2.2). On $[a, \infty)$, a > 0, it is uniform: gap $\leq \frac{\pi}{2} - \arctan(na) \to 0$.

In the model paper: **Q15** **Q20** **Q23** (PYQ MAT-22 Q31) **Q30**.

# T15 Sums and products

Bartle Exercises 8.1.21–23 (PYQ MTS-22 MCQ 13).

**RESULT**

If $f_n \to f$ and $g_n \to g$ uniformly on A, then $f_n + g_n \to f + g$ and $cf_n \to cf$ uniformly. If also every $f_n$ and $g_n$ is **bounded**, then $f_n g_n \to fg$ uniformly.

**PROOF IDEA**

**Sum:** $|(f_n + g_n) - (f + g)| \le |f_n - f| + |g_n - g|$, so $\|\cdot\| \le \|f_n - f\| + \|g_n - g\| \to 0$.

**Product:** $f_n g_n - fg = f_n(g_n - g) + g(f_n - f)$. With $\|f_n\| \le M$ for all large n, $\|f_n g_n - fg\| \le M\|g_n - g\| + \|g\|\,\|f_n - f\| \to 0$.

**WITHOUT BOUNDEDNESS, PRODUCTS FAIL**

$f_n = x + \dfrac{1}{n} \to x$ uniformly on $\mathbb{R}$ (norm 1/n). But $f_n^2 - x^2 = \dfrac{2x}{n} + \dfrac{1}{n^2}$; at $x = n$ this is $\ge 2$. So $f_n^2 \not\to x^2$ uniformly on $\mathbb{R}$ C23 .

In the model paper: Q12 (PYQ MTS-22 MCQ 13), Q29 , Q36 .

# Counterexample library

> Each one breaks a tempting wrong idea. Learn the sequence, the set and the one-line reason.

| CODE | SEQUENCE, SET | WHAT IT SHOWS |
|---|---|---|
| C12 | $x^n$ on [0, 1] | continuous $f_n$, pointwise limit jumps at 1; norm of gap = 1 |
| C13 | triangle spikes, height n, base 2/n | $f_n \to 0$ but $\int f_n = 1$ for all n |
| C15 | $x/n$ on $\mathbb{R}$ | pointwise, not uniform; uniform on every bounded set |
| C21 | $nx/(1+n^2x^2)$ on $[0,\infty)$ | moving bump of fixed height ½: not uniform, continuous limit |
| C22 | $x^n/(1+x^n)$ on [0, 2] | norm of gap exactly ½, never attained |
| C23 | $x + \frac{1}{n}$ on $\mathbb{R}$ | uniform, but its square is not: products need boundedness |
| C24 | $\arctan(nx)$ on $\mathbb{R}$ | limit $\frac{\pi}{2}\operatorname{sgn} x$ jumps at 0; uniform on $[a,\infty)$ |
| C25 | $x^n$ on [0, 1) | continuous limit 0, still **not** uniform |
| C26 | $nx/(1+nx)$ on [0, 1] | not uniform, yet $\int f_n \to \int f$ |
| C27 | $n^2x^2e^{-nx}$ on $[0,\infty)$ | peak $4/e^2$ at $x = 2/n$ for every n: not uniform |

**YOUR INTUITION WILL LIE TO YOU HERE**

"Every $f_n$ is close to 0 at each point, so the graphs are close to 0." C21 and C27: every point settles, but a bump of fixed height keeps moving. Pointwise is about points; uniform is about graphs.

# Which tool?

> Mixed drill. Name the tool first, then check its conditions. Answers below.

1. Find $\lim_n f_n(x)$.
2. Show $f_n \to f$ uniformly, and $|f_n(x) - f(x)| \leq$ something with no x.
3. Show uniform, but the gap depends on x.
4. Show not uniform; the $f_n$ are continuous and the limit jumps.
5. Show not uniform; the limit is continuous.
6. Show uniform convergence without knowing f.
7. Prove a uniform limit of continuous functions is continuous.
8. Show $\lim \int f_n \neq \int \lim f_n$.

**ANSWERS**

1. Fix x, cases, P1–P2 (T2). 2. Bound free of x (T8). 3. Norm by calculus, Lemma 8.1.8 (T6, T7). 4. Theorem 8.2.2 contrapositive (T14). 5. Points $x_k$ with gap ≥ $\varepsilon_0$ (Lemma 8.1.5, T4), or norm not → 0. 6. Cauchy criterion 8.1.10 (T10). 7. ε/3 argument (T13). 8. Compute both sides (T12).

# Theorem sheet

**8.1.1 POINTWISE**

for each $x \in A_0$, $f_n(x) \to f(x)$; K may depend on ε and x.

**8.1.4 UNIFORM**

for each ε there is K(ε), not depending on x, with $|f_n(x) - f(x)| < \varepsilon$ for all $n \geq K$ and all $x \in A_0$.

**8.1.5 NOT UNIFORM**

⇔ some $\varepsilon_0 > 0$, subsequence $f_{n_k}$, points $x_k$ with $|f_{n_k}(x_k) - f(x_k)| \geq \varepsilon_0$.

**8.1.7 UNIFORM NORM**

$\|\varphi\|_A = \sup\{|\varphi(x)| : x \in A\}$ for bounded φ.

**8.1.8 NORM TEST**

bounded $f_n$: uniform on A ⇔ $\|f_n - f\|_A \to 0$.

**8.1.10 CAUCHY CRITERION**

bounded $f_n$: uniform to a bounded f ⇔ ∀ε ∃H: $\|f_m - f_n\|_A \leq \varepsilon$ for $m, n \geq H$.

**8.2.2 CONTINUITY**

$f_n$ continuous on A, $f_n \to f$ uniformly on A ⇒ f continuous on A.

# Recall plan

> Testing yourself beats re-reading. Tick each box.

**DAY 0 (TODAY)**

☐ Read T1–T15 with a pencil. Cover each proof and explain every "why" line aloud.

**DAY 1**

☐ From memory, write Definitions 8.1.1, 8.1.4, 8.1.7. Check them word for word against this file.

**DAY 3**

☐ Without looking, write the proof of Theorem 8.2.2 from its skeleton. ☐ Do the $x^n(1-x)$ norm.

**DAY 7**

☐ Do the "Which tool?" drill. ☐ Write the Cauchy criterion proof.

**DAY 14 / EXAM WEEK**

☐ Say every counterexample in the library aloud with its reason. ☐ Do the model paper, timed.

## How to attack a question

| | |
|---|---|
| **3 marks** | Exact definition or statement, or one example with a one-line check. About 4 minutes. |
| **6 marks** | Pointwise limit, then the gap, then its sup (or $x_k$), then the verdict with the theorem named. 8–12 lines. |
| **10 marks** | Statement word for word, Know/Want, numbered proof, then "where each hypothesis was used". If stuck, still write the skeleton. |

# Past questions → topics

MTS-25 = MTS 6B 10, March 2025 · MTS-26 = April 2026 · MTS-22 = March 2022 (and its MCQ paper) · MAT-22 = MAT 6B 09, March 2022 · MAT-21 = MAT 6B 09, March 2021. These are papers of the old CBCSS course with the same textbook; the FYUGP course MAT5CJ301 has no papers yet.

| PAST QUESTION | TOPIC | MODEL Q |
|---|---|---|
| MTS-25 Q9 uniform norm | T5 | Q1 |
| MTS-25 Q10 $\lim e^{-nx}$ | T2 | Q4 |
| MTS-25 Q20 uniform limit continuous | T13 | Q38 |
| MTS-25 Q26 Cauchy criterion | T10 | Q37 |
| MTS-26 Q9 define uniform | T3 | Q3 |
| MTS-26 Q10 $\lim nx/(1+nx)$ | T2 | Q5 |
| MTS-26 Q20 norm test | T6 | Q39 |
| MTS-22 Q8 define pointwise | T1 | Q2 |
| MTS-22 Q9 $x^n$ on (−1, 1] | T4 | Q9 |
| MTS-22 Q10 $2nxe^{-nx^2}$ integrals | T12 | Q10 |
| MTS-22 MCQ 11 $x^n$ continuity | T11 | Q11 |
| MTS-22 MCQ 12 $\sin(nx+n)/n$ | T8 | Q8 |
| MTS-22 MCQ 13 $2f_n, f_n^2, f_n^3$ | T15 | Q12 |
| MAT-22 Q8 $\lim(x^2+nx)/n$ | T2 | Q6 |
| MAT-22 Q19 norm test (⇐) | T6 | Q39 |
| MAT-22 Q31 $x^n/(x^n+1)$ on [0, 2] | T14 | Q23 |
| MAT-21 Q10 $\lim\cos(nx+n)/n$ | T2 | Q7 |
| MAT-21 Q21 pointwise, test $x^n$ | T2 | Q21 |
| MAT-21 Q32 $x^n(1-x)$ | T7 | Q22 |
| MAT-21 Q37b $x^n$ on [0, 1] | T9 | Q40 |

| | | |
|---|---|---|
| MAT-21 Q38a Cauchy criterion | T10 | Q37 |

Left out (series of functions, Weierstrass M-test, Theorem 8.2.4: outside the MAT5CJ301 Module 4 list): MTS-25 Q11, MTS-26 Q11 and Q26, MTS-22 Q11, MAT-22 Q7, Q20, Q21, Q22, MAT-21 Q9, Q23, Q31.